

IP	Internet protocol
TCP	Transmission control protocol
UDP	User datagram protocol
PVC	Permanent virtual circuit
SVC	Switched virtual circuit
ATM	Asynchronous transfer mode
CBR	Constant Bit Rate
VBR	Variable Bit Rate
ABR	Available Bit Rate
VPN	Virtual Private network
VPI	virtual path indicator
VCI	virtual circuit indicator
CIDR	Classless Inter-domain Routing
ARP	Address Resolution protocol
MTU	Maximum Transmission Unit
ICMP	Internet Control message protocol
NAT	Network Address translation
NAPT	Network address & port
IGPs	Interior Gateway protocols
EGPs	Exterior Gateway protocols
BGP	Border Gateway protocol
RIP	Routing Information protocol
OSPF	Open shortest path 1st protocol

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- IGMP Internet Group multicast protocol
- RPB Reverse path Broadcasting
- DVMRP Distance vector multicast Routing protocol
- CBT Core Based trees
- PIM-SM protocol Independent multicast-Sparse mode
- PIM-DM protocol Independent multicast-Dense mode
- MOSPFF Multicast extensions To the Open shortest path 1st protocol

ATM QoS Specifications

CBR VBR ABR

Layering of Tcp/IP protocols

- 1 Physical = Basic network hardware
- 2 Network Interface = organize data in frames
- 3 Internet = Specifies the format of packets sent across an Internet and forward packets
- 4 Transport = ensure reliable transfer
- 5 Application = How application uses an Internet

Address Resolution Techniques

- 1) Table Lookup
- 2) Closed-form Computation
- 3) Message exchange

Header Information (Datagram)

- 1) Destination address
- 2) Source address
- 3) Identifier
- 4) other delivery information

Advantages of multiple headers

- o) Efficiency
- o) Flexibility
- o) Incremental development

Types of Addresses

- o) Unicast
- o) multicast
- o) Cluster

Internet layer Detect errors

- o) checksum (header only)
- o) TTL expires
- o) No route destination network
- o) can't deliver to destination host

Error Messages

- o) Source Quench
- o) Time exceeded
- o) Destination unreachable
- o) Redirect
- o) Fragmentation Required

Informational messages

- o) echo request / reply
- o) Address mask request / reply
- o) Router discovery

Internet Routing protocols

- o) Border Gateway protocol
- o) Routing Information protocol
- o) open shortest path 1st protocol

Classification of IRP

- o) Interior Gateway protocol
- o) Exterior Gateway protocol.